



Samuel D. Pollard

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Overview

I am a senior member of technical staff in cybersecurity researching formal verification at Sandia National Laboratories. My research focuses on the application of formal methods to high performance computing, embedded system verification, and floating-point arithmetic.

Education

Ph.D. computer science, *University of Oregon*: 2016–2021
Advisor: Boyana Norris. GPA: 4.07.
Dissertation: *When Does a Bit Matter? Techniques for Verifying the Correctness of Assembly Languages and Floating-Point Programs*. [PDF]

M.S. computer science, *Western Washington University*: 2014–2016
Supported by teaching and research assistantships. GPA: 3.95.

B.S. mathematics, *Western Washington University*: 2010–2014
Graduated with university honors, computer science minor. GPA: 3.60.

Employment

Senior R&D scientist, cybersecurity, Sandia National Laboratories: 2021–Present
Roles include: researcher, principal investigator, subject matter expert, and software engineer, all with a focus on formal verification of high-consequence systems.
Principal investigatory roles include three Laboratory-Directed Research & Development Projects (LDRD), a DARPA formal methods project, and a sub-element role for NNSA Advanced Simulation and Computing and University contracts
Project lead for portions of 8 Sandia scientists and 2 university collaborations; managing more \$1.5 million total budget per year.
Intern mentoring: Haoda “Harry” Wang (Columbia), John Jacobson III (U. of Utah), Kaden Hart (Utah State U.), Sabrina Reis (U. Oregon), Anthony Dario (U. Oregon), Laine Rumreich (The Ohio State U.), Ariel Kellison (Cornell), Jarom Christiansen (The Ohio State U.), Alex Zhang (UIUC), Noam Steiner Tomer (North-eastern), Francisco Vigil (U. Michigan), Jackson Brough (U. Utah)

Formal Methods R&D Intern, Sandia National Laboratories: 2018–2021
Year-round intern mentored by Geoff Hulet and Philip Johnson-Freyd.
Developed a Haskell-based tool for binary analysis of legacy architecture [9].

Computation Student Intern, Lawrence Livermore National Laboratory: 2017
Summer internship researching topology-aware job scheduling [2].

Software Engineering Intern, EMC Isilon: 2015
Developed of internal testing utilities to ensure operating system reliability.

Teaching Assistant, Western Washington University: 2014–2015
Responsible for instructing and grading four lab sections (approximately 20 students each) of 100 and 200-level computer science courses.

Publications

Conferences
[1] Samuel D. Pollard, Sudharshan Srinivasan, and Boyana Norris. A performance and recommendation system for parallel graph processing implementations: Work-in-progress. In *Companion of the 10th ACM/SPEC International Conference on Performance Engineering*, ICPE ’19, pages 25–28, Mumbai, India, April 2019. ACM.

- [2] Samuel D. Pollard, Nikhil Jain, Stephen Herbein, and Abhinav Bhatele. Evaluation of an interference-free node allocation policy on fat-tree clusters. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage, and Analysis*, SC '18, pages 26:1–26:13, Dallas, TX, USA, November 2018. IEEE Press.

Journals

- [3] Sriram Srinivasan, Samuel D. Pollard, Sajal K. Das, Boyana Norris, and Sanjukta Bhowmick. A shared-memory algorithm for updating tree-based properties of large dynamic networks. *IEEE Transactions on Big Data*, pages 1–15, September 2018.

Workshops

- [4] Max Fan, Ariel E. Kellison, and Samuel D. Pollard. Mechanizing olver’s error arithmetic. In *International Workshop on Verification of Scientific Software*, VSS, May 2025.
- [5] Anthony Dario and Samuel D. Pollard. A step-function abstract domain for granular floating-point error analysis. In *Proceedings of the 10th ACM SIGPLAN International Workshop on Numerical and Symbolic Abstract Domains*, NSAD '24, New York, NY, USA, October 2024. ACM.
- [6] Zachary Sullivan and Samuel D. Pollard. A formal model for portable, heterogeneous accelerator programming. In *Vivek Sarkar Festschrift Symposium*, VIVEKFEST, October 2024.
- [7] Samuel D. Pollard, Robert C. Armstrong, John Bender, Geoffrey C. Hulette, Raheel S. Mahmood, Karla Morris, Blake C. Rawlings, and Jon M. Aytac. Q: A sound verification framework for statecharts and their implementations. In *8th International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS)*, pages 16–26, New York, USA, December 2022. ACM.
- [8] Samuel D. Pollard and Boyana Norris. A statistical analysis of error in MPI reduction operations. In *Fourth International Workshop on Software Correctness for HPC Applications*, Correctness, pages 49–57. IEEE, November 2020.
- [9] Samuel D. Pollard, Philip Johnson-Freyd, Jon Aytac, Tristan Duckworth, Michael J. Carson, Geoffrey C. Hulette, and Christopher B. Harrison. Quameleon: A lifter and intermediate language for binary analysis. In *Workshop on Instruction Set Architecture Specification*, SpISA '19, pages 1–4, Portland, OR, USA, September 2019.

Posters

- [10] Vivek Kale, Shyamali Mukherjee, Samuel D. Pollard, Richard Rutledge, and Jackson Mayo. Automated analysis of heterogeneous parallel kokkos applications guided by example programs. Poster in 1st Annual Conference of the US Research Software Engineer Association (US-RSE'23), October 2023.
- [11] Samuel D. Pollard and Boyana Norris. A comparison of parallel graph processing implementations. In *IEEE International Conference on Cluster Computing*, CLUSTER, pages 657–658, Honolulu, HI, USA, September 2017. IEEE Computer Society.

Technical Reports and Position Papers (non-refereed)

- [12] Samuel D. Pollard, Jon M. Aytac, Ariel Kellison, Ignacio Laguna, Srinivas Nedunuri, Sabrina Reis, Matthew J. Sottile, and Heidi K. Thornquist. The first tri-lab workshop on formal verification: Capabilities, challenges, research opportunities, and exemplars. Technical report, Sandia National Laboratories, February 2024.
- [13] Ariel Kellison, Geoff C. Hulette, John Bender, Samuel D. Pollard, and Heidi K. Thornquist. Formal methods-based certification frameworks for scientific computing applications. Technical report, ASCR Workshop on Cybersecurity and Privacy for Scientific Computing Ecosystems, U.S. Department of Energy, Office of Advanced Scientific Computing Research, November 2021.
- [14] Noah Evans, Robert Armstrong, Samuel D. Pollard, and Jacob Hobbs. AI-based formal specification for scientific security. Technical report, ASCR Workshop on Cy-

bersecurity and Privacy for Scientific Computing Ecosystems, U.S. Department of Energy, Office of Advanced Scientific Computing Research, November 2021.

Service

Peer Review

Program committee: Supercomputing Correctness Workshop (2022–2026), Vivek Sarkar Festschrift Symposium (2024), Verification of Scientific Software (2025).
Alliance Strategy Team: Predictive Science Academic Alliance Program IV (2024–2026).
Committee member: Halpewattage Akila Nuwantha Peiris, *Dungeons and Dragons Fantasy Adventure Generation*, M.S. Thesis, University of Moratuwa, 2024.
Reviewer: Supercomputing Correctness Workshop 2018–2021, ACM Transactions on Mathematical Software 2019.

Mentoring

Panelist, ACM PLMW@SPLASH Career Paths, Pasadena, CA, 2024.
ACM SIPLAN-M long-term mentoring, 2023–2025.
Early Career Professional Mentor (NSF Grant #1742110), Western Washington University: 2018–2020.
ACM tutor, Western Washington University, 2013–2014.

Graduate Student Union Steward: 2019–2021

Point of contact between computer science graduate students and union leadership.

Supercomputing Student Volunteer, ACM/IEEE: 2018

Student Representative, Graduate Education Committee, Computer and Information Sciences, University of Oregon: 2017–2019
Provided student perspectives on department policies.

Radio Station DJ: 2010–2011

Disc jockeyed for KUGS, Western Washington University's radio station.

Honors & Awards

Sandia National Labs Employee Recognition Awards

Lab news article published on my work in floating-point error analysis, [Fixing FLOPS]
2025: Collaboration team award for work on the Exploratory Signal Processing Platform project.
2022: Technical excellence team award for work on formally-verified compilation.

General University Scholarship, University of Oregon: 2019 & 2020

Awarded based on scholastic achievement and service to the university.

ICPE 2019 Travel Grant: April 2019

The 10th ACM/SPEC International Conference on Performance Engineering.

IEEE Student Mentor Program: September 2017

Travel grant to attend IEEE Cluster '17 funded by NSF and IEEE.

Erwin & Gertrude Juilfs Scholarship in Computer and Information Science, John Juilfs: 2017

Scholarship awarded by nomination from the departmental committee at the University of Oregon.

Kaiser-Borsari Scholarship, Kaiser-Borsari Educational Foundation: 2015

Scholarship awarded to students who demonstrate high academic potential combined with creativity and leadership qualities and financial need.

COMAP Mathematical Contest in Modeling, Western Washington University: 2013
Group project in statistical modeling. Received meritorious (among the top 9%).

Mathematics Memorial Scholarship, Western Washington University: 2013
Tuition waiver awarded based on grade point averages in the Dept. of Mathematics.

Washington State Opportunity Scholarship, College Success Foundation: 2012 & 2013
Merit-based scholarship for low-income STEM students in Washington State.

Presidential Scholarship, Western Washington University: 2010
Two year scholarship awarded to the top 1% of applicants to WWU.

Washington State PTA Scholarship, Washington Congress of Parents and Teachers: 2010
Merit-based scholarship awarded to seniors from WA State public high schools.

Federal Pell Grant, U.S. Department of Education: 2010–2014
Awarded to undergraduate students who display exceptional financial need.

Skills

Formal Methods: Rocq, type theory, automated reasoning, probabilistic programming languages, floating-point error analysis.

Software Engineering: Git, Subversion, Make/CMake, Bash, R, MATLAB, C++, C, Python, Bash, Awk, OCaml, Ada, Haskell, Java, Perl, Icon, Racket, Fortran, x86-64 assembly, $\text{\LaTeX}$ /Tikz.

Mathematics: Numerical analysis, abstract algebra, probability, statistics, topology, and real analysis.

Hobbies

My hobbies include homebrewing, creative writing, jewelry, cycling, computer generated art, painting, and video gaming (mostly Dota 2).